

Module 04 (c)

Text Search with grep

Fundamentals & Basic Syntax

Understanding pattern matching in plain text
files

What is grep?

Origin & Acronym

Derived from the early Unix text editor `ed` command `g/re/p`:
Globally search for a **R**egular **E**xpression and **P**rint matching lines.

Primary Function

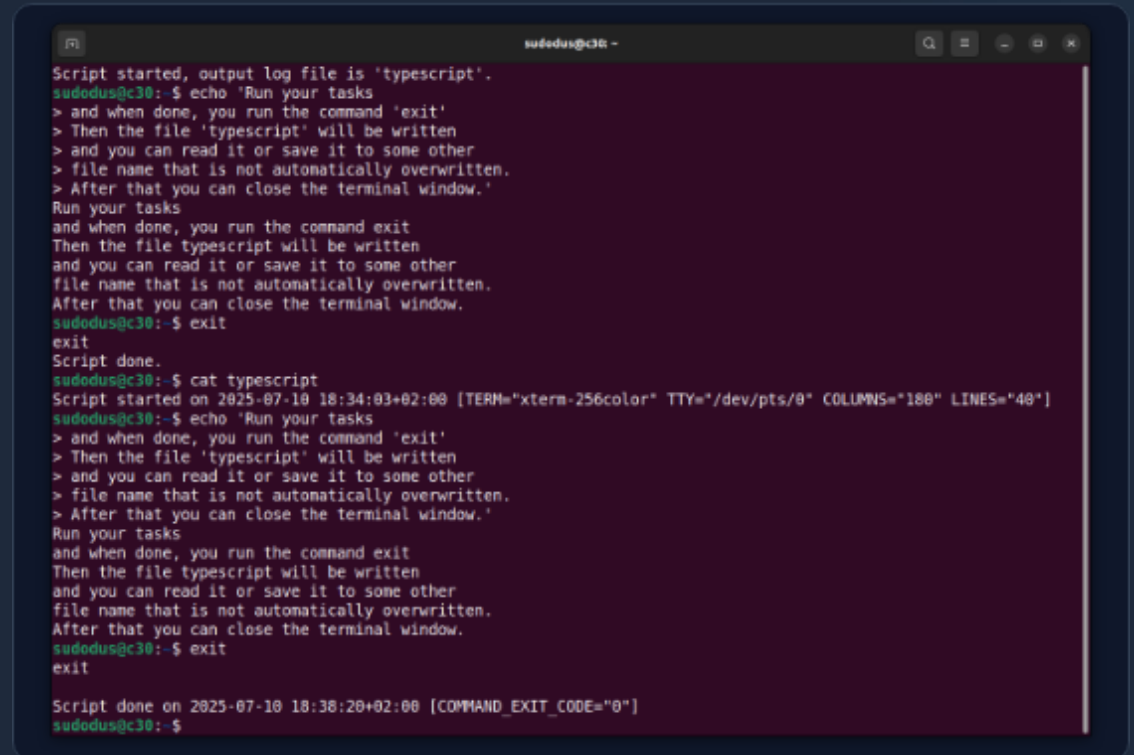
Scans input files line-by-line, searching for lines that match a specified pattern, and writes matching lines directly to standard output (stdout).

Basic Syntax & Usage

The standard syntax for invoking grep on a Linux terminal:

grep [options] "pattern" filename

- **pattern:** The text or regular expression to search for.
- **filename:** Target file or path to search within.
- **Quoting:** Always quote patterns to prevent shell expansion.



```
Script started, output log file is 'typescript'.
sudodus@c30: $ echo 'Run your tasks
> and when done, you run the command 'exit'
> Then the file 'typescript' will be written
> and you can read it or save it to some other
> file name that is not automatically overwritten.
> After that you can close the terminal window.'
Run your tasks
and when done, you run the command exit
Then the file typescript will be written
and you can read it or save it to some other
file name that is not automatically overwritten.
After that you can close the terminal window.
sudodus@c30: $ exit
exit
Script done.
sudodus@c30: $ cat typescript
Script started on 2025-07-10 18:34:03+02:00 [TERM="xterm-256color" TTY="/dev/pts/0" COLUMNS="180" LINES="48"]
sudodus@c30: $ echo 'Run your tasks
> and when done, you run the command 'exit'
> Then the file 'typescript' will be written
> and you can read it or save it to some other
> file name that is not automatically overwritten.
> After that you can close the terminal window.'
Run your tasks
and when done, you run the command exit
Then the file typescript will be written
and you can read it or save it to some other
file name that is not automatically overwritten.
After that you can close the terminal window.
sudodus@c30: $ exit
exit

Script done on 2025-07-10 18:38:20+02:00 [COMMAND_EXIT_CODE="0"]
sudodus@c30: $
```

Essential Command Line Flags



Case Insensitive (-i)

Ignores letter case distinctions during pattern matching.

```
grep -i "error" log.txt
```



Line Numbers (-n)

Prefixes each matching line with its 1-based line number in the source file.

```
grep -n "main" main.c
```



Invert Match (-v)

Inverts the search to select and output all non-matching lines.

```
grep -v "DEBUG" app.log
```

Recursive Directory Search



Recursive Flag

Searching Across Entire Codebases

Use `grep -r "pattern" dir/` to search through all files in a directory hierarchy recursively.

Practical Example:

```
grep -rn "pthread_create" ./src/
```

Finds all C source threads initialization calls recursively with line numbers.

Quick Reference: Common Flags

Flag	Name	Description	Example Usage
-c	Count	Prints only the total count of matching lines per file	grep -c "FAIL" test.log
-w	Word Match	Matches whole words only (enforces word boundaries)	grep -w "int" program.c
-l	List Files	Displays only the names of files containing matches	grep -rl "TODO" src/
-E	Extended Regex	Enables Extended Regular Expressions (ERE) syntax	grep -E "err warn" log.txt

Regex Anchors & Context Control

Leveraging regular expressions for precise text
filtration

Regular Expression Anchors

- </> Start of Line (^):** Matches patterns starting at line origin. `grep "^#include" file.c` extracts header includes.
- </> End of Line (\$):** Matches patterns ending a line. `grep "return 0;$" file.c` finds clean main function terminations.
- </> Empty Lines (^\$):** Combining both anchors matches empty lines. Use `grep -v "^$" file.txt` to strip blank lines.
- </> Character Classes ([]):** Matches any single character listed. `grep "[0-9]" data.txt` finds lines containing digits.

Context Flags: -A, -B, -C

-A N: Shows N lines *After* the match.

-B N: Shows N lines *Before* the match.

-C N: Shows N lines of *Context* (both directions).

```
grep -C 2 "panic" kernel.log
```

Grep Variants Performance Comparison

fgrep (-F) Fastest (Fixed Strings)

grep (BRE) Standard (Basic Regex)

egrep (-E) Flexible (Extended Regex)

Note: fgrep bypasses regex parsing engines entirely, achieving maximum throughput when searching fixed text literals in large dataset logs.

Questions & Lab Practice

Try running `man grep` on your Linux terminal!

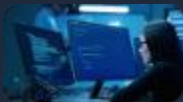
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Image Sources



<https://i.sstatic.net/LqX1frdr.png>

Source: askubuntu.com



https://static.vecteezy.com/system/resources/thumbnails/081/654/582/small_2x/young-asian-female-programmer-writes-code-on-multiple-monitors-in-a-dark-high-tech-cybersecurity-center-it-specialist-analyzes-software-data-late-at-night-modern-tech-office-with-a-male-colleague-video.jpg

Source: www.vecteezy.com